

Total no. of Pages:

Roll no.

Degree: B.Tech (ECAM)

Semester: Third semester

MID-SEMESTER EXAMINATION, September, 2024

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Course Code: EAEPC08

Duration: 1:30 Hours

Max. Marks: 15

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Given octal numbers $X = (263)_8$ and $Y = (157)_8$. Find their binary and decimal values.	1.5	1
1b	If $P = 574$ and $Q = 683$, find the BCD sum of P and Q.	1.5	1
2a	Given that $x = C8 H$ and $y = 7D H$, evaluate $(x - y)$ using 2's complement.	1.5	1
2b	$F(A, B, C, D) = \sum \text{min} (1, 6, 8, 10, 14) + d(0, 2, 9)$; Minimize using K-map.	1.5	1
3a	Design a 2-bit comparator logic circuit	1.5	1, 2
3b	Draw a block diagram of a single digit BCD adder	1.5	2
4a	Implement using a 3 to 1 Multiplexer. $F(A, B, C, D) = \sum \text{min} (0, 1, 3, 4, 5, 8, 10, 12, 14, 15)$	1.5	2
4b	Design a Full Adder using a De-multiplexer	1.5	2
5a	Explain the following characteristics of TTL logic family: Propagation delay and (ii) Noise margin	1.5	2
5b	Design a CMOS circuit to implement: $F(w, x, y) = (w + x.y + w.y)'$	1.5	2